

Course Number	0418-416	Course Title	Modeling and Performance Evaluation
Credit Hours	3 (3:0:3)	Course Coordinator	Dr. Hosam AboElfotoh
Compulsory	No	Prerequisite	0418-331

Textbook

<i>Title:</i>	<i>The Art of Computer Systems Performance Analysis</i>
<i>Author:</i>	<i>Raj Jain</i>
<i>Publisher:</i>	<i>John Wiley & Sons, Latest Edition</i>

Major Topics Covered in the Course

	Topics	Hours
1	Introduction: Modeling and simulation; performance measures for computer systems.	5
2	Background: Probability theory, random variables, distributions, stochastic processes, means, variances, correlations.	8
3	Simulation techniques: Random number generation, generating random variates; programming simulations; time-based and event-based simulation; accumulating statistics, analyzing simulation results.	9
4	Stochastic processes: Classification; single queues, network of queues; Markovian systems.	9
5	Case studies: Time-sharing systems, client-server systems, local area networks.	8
6	Midterm I and Midterm II.	3
	Total.	42

Grading

15-20%	Midterm 1
15-20%	Midterm 2
30-20%	Hmk/ quiz/project
40%	Final

Course Outcomes

Students should be able to:

1. Demonstrate the ability to model and simulate different types of computer systems and networks.
2. Demonstrate the ability to apply the techniques of modeling and simulation to solve problems related to performance.
3. Use computer simulation to evaluate and predict the performance of computer systems and networks.

Relationship between Course Outcomes and Student Outcomes

Course Outcomes	Student Outcomes					
	1	2	3	4	5	6
1		H				M
2	H	M				M
3		M				